Phase 1: AI Project Proposal
Email Content Summarization & Scheduler

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Project topic: Students struggle to schedule their timetables well through AIMS.
Target users/stakeholders: Students using AIMS, particularly students with large flexibility in course scheduling.

1 Problem Definition & Scope

1.1 Problem

The university students will receive a large volume of emails everyday, which contains assignment / quiz / exam announcement, event invitation, security alert, bank statement notification, promotion, etc. Spending time on checking these emails can be annoying. And the performance of filter function by email apps like Gmail and Outlook is not good enough. After checking these emails, make it to the schedule can also be another issues to the university students.

1.2 Scope

The project will contains classification of the emails, summarize an email from paragraph form to bullet-point form, integrate different emails to generate a schedule. But we will make an agent to help users to write emails.

2 AI Task Formulation

Task: Rank a set of timetables based on selected selection criteria, including, but not necessarily limited to, time of day, day of week, and consecutiveness

Inputs: List of occupied schedule slots / Slot in schedule's status as occupied / unoccupied (6 days of the week × 13 possible hours per day = 78 binary inputs)

Outputs: Scores based on various criteria (eg: "earliness", "lateness", "consecutiveness", etc; one output per criteria, potentially with an additional final layer for an overall aggregate score)

Constraints: Scores outputted from 0-100. Model will only be fed valid timetables (ie. without collisions)

Evaluation: Judge accuracy of suggestions; Scores should agree with the opinion of users.

Data source: Survey students for their opinion on how a set of generated timetables should be scored. No privacy, consent, or access considerations needed.

3 Proposed AI Approaches

Present multiple plausible AI approaches. For each approach, briefly explain the core method, required data/resources, expected workflow, and practical feasibility. Divergent solutions are encouraged; do not present only one method under different names.

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3.1 AI Agent

build an agent with several tools, including a email reader, summarizer, report generator and scheduler. The agent will read emails, summarize them, and schedule tasks based on the summarized content. And the task can be timely triggered by the agent based on the email content. The agent will be able to learn from user feedback and improve its performance over time.

3.2 Reinforcement Learning

The agent will treat the received emails as a environment and the actions taken by the agent as a policy. The function

3.3 naive bayes

Use a naive bayes classifier to classify the emails into different categories based on the words frequence and the context of the email. The classifier will first be trained on a labeled dataset of emails and then used to classify new emails.

3.4 decision tree

Use a decision tree classifier to classify the emails into different categories based on the words frequence and the context of the email. The classifier will first be trained on a labeled dataset of emails and then used to classify new emails.

3.5 hard-code Rule-based pipeline

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3.6 Convolutional NN

- Benefits
  - Better at preserving adjacency data
    - More accurate scores regarding “adjacency”
- Weaknesses
  - (to be written)
- Core method
  - Run a convolution kernel across the squares of the timetable before feeding the results through a simple fully connected neural network.
- Required resources
  - A laptop running Pytorch
- Required data
  - A set of randomly generated schedules and scores assigned by volunteers.

3.7 Linear regression

- Benefits
  - Trivial to optimize.
- Weaknesses
  - (to be written)
- Core method
  - (to be written)
- Required resources
  - (to be written)
- Required data
  - (to be written)

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4 Comparative Analysis

I think AI agent method is the best.

5 Expected Impact & Project Feasibility

Expected benefit: Easier scheduling, more efficient time management

Novelty: Likely very novel?

Value: High, helps students and the university schedule more efficiently, balance the preferences of students and professors

Data: Data of courses (scraped from AIMS Master Timetable), user preference data (for which timetables are scored higher)

Data collection plan: Survey students to select preferred timetables from generated timetables or score generated timetables. (eg. Ask students to score a set of 5 timetables from 0-100 based on the set of criteria listed above.)

Implementation: Create a dataset with scored timetables and optimise the regression/convolutional model to assign appropriate scores to schedules.

Tools: uhhh

Risks: Extremely minor: No identifying information needs to be collected (eg. name, class, major) and all collected information is not

Mitigation: N/A

Access to hardware: For Linear regression: Trivial (one laptop)

Access to software:

Access to data:

References

[1] Lance Le, "A Sample Title," *A sample journal*, 2023.

[2] Author Maple, "Rubik's Cube Methods," *Journal of Cube*, 2021.